Step-by-Step Project Development Procedure (2010–2026 Dataset)

Assigned To: Jiyansh Gokhru | **Dataset:** F1 Racing Project Dataset.xlsx (5 Sheets • 13,748 Rows) | **Visualizations:** 9 Unique Charts | **Story:** 9 Narrative Scenes

1. Problem Statement & Multi-Sheet Dataset Architecture (To-Do 1 & To-Do 2)

Formula 1 racing generates high-volume race classification, qualifying, and lap-by-lap telemetry records. The objective of this project is to connect the multi-sheet Excel dataset (**F1 Racing Project Dataset.xlsx**, spanning **17 seasons from 2010 to 2026** and **13,748 combined rows**) into Tableau Desktop and a Flask web application, perform rigorous data preparation, build **9 unique interactive visualizations** inside a unified dark-themed dashboard, and present a **9-scene data story**.

Sheet Name	Rows	Cols	Primary Join Keys	Schema & Analytical Role
race_results	6,464	42	race_id, circuit_id	Primary race outcomes, grid/finish positions, points, 396 DNFs with dnf_reason, pit stops, starting tyre compound, weather, safety cars, DRS overtakes, laps led
lap_intelligence	3,814	14	race_id, circuit_id	Lap-by-lap telemetry (lap_time_s, sector1_s, sector2_s, sector3_s), tyre_compound, tyre_age_laps, drs_active, speed_trap_kmh, is_pit_lap
qualifying_results	3,130	16	race_id, circuit_id	Qualifying times (q1_lap_time_s, q2_lap_time_s, q3_lap_time_s), gap_to_pole_s, sector times, top_speed_kmh
championship_standings	311	10	season, driver, team	Season classifications, points, wins, podiums, pole_positions, dnfs, champion flag
circuits	29	9	circuit_id	29 global circuits across 21 countries (circuit_name, country, city, laps, lap_km, type, altitude_m, direction)

2. Prepare the Data for Visualization (To-Do 3, To-Do 7, To-Do 8, To-Do 9)

Preparation Step	Implementation Details on Provided Dataset
Null & Missing Value Treatment	Records in race_results with missing finish_position (396 DNF entries) were retained with the dnf_reason column populated (Engine: 73, Hydraulics: 70, Mechanical: 67, Brake failure: 64, Gearbox: 62, Collision: 60) to support Driver Reliability Analysis. Null gap_to_leader_s values on DNF rows were excluded from gap-based aggregations.
Data Type Validation	Numeric measures (points, laps_led, drs_overtakes, air_temp_celsius, lap_time_s) were confirmed as continuous fields. Categorical dimensions (driver, team, country, circuit_type, tyre_compound, dnf_reason) were confirmed as discrete dimensions.
Calculated Fields Created	A DNF Count field was derived from the dnf flag to count total DNFs per driver across all seasons. A Points Cumulative field was computed as a running total of points per driver per season.
Parameter Setup	An integer 'Top N Parameter' (default value: 10, range 1–19) was configured to dynamically control how many drivers appear on the Driver Performance Leaderboard via a calculated filter.
Logical Layer Relationships	race_results was joined to circuits on circuit_id and to lap_intelligence & qualifying_results on race_id. championship_standings was kept as an independent sheet for Team Championship Performance Analysis.

Preparation Step	Implementation Details on Provided Dataset
Global Filter Configuration	DNF Reason, Year of Race Date (2010–2026), and Top N Parameter were configured as dashboard-level global controls in Extract mode.

3. The 9 Unique Dashboard Visualizations & Exact Results (To-Do 4, To-Do 5, To-Do 10)

Visualization Title	Chart Type	Verified Dataset Metrics & Key Findings
1. KPI Summary Cards (4 Tiles)	KPI Cards	Competing Teams: 10 | Seasons in Play: 17 (2010–2026) | Race Outcomes Logged: 6,464 | Circuits Showcased: 29
2. Global Driver Distribution Across Grand Prix Nations	Geographic Map	21 Host Nations: Bahrain leads (2,168 entries across 115 races), followed by USA (400), Monaco (328), Hungary (328), Spain (328), Canada (308), UAE (308), Brazil (308), Singapore (288), Italy (220), Austria (180), Mexico (160).
3. Driver Reliability Analysis	Stacked Vertical Bar Chart	396 Total DNFs stacked by dnf_reason (Engine: 73, Hydraulics: 70, Mechanical: 67, Brake failure: 64, Gearbox: 62, Collision: 60). Highest DNFs: Carlos Sainz (34), Sergio Perez (29), Charles Leclerc (28). Best reliability: Lewis Hamilton (12 DNFs).
4. Driver Performance Leaderboard	Vertical Bar Chart (Top N)	Controlled by Top N Parameter (default 10). Direct Points: Max Verstappen (4,640 pts, 100 wins), Lewis Hamilton (4,491 pts, 118 wins), Sergio Perez (3,366), George Russell (3,299), Charles Leclerc (3,243), Carlos Sainz (3,147), Lando Norris (2,454). (Logical-layer circuit points: Hamilton 31,147; Verstappen 31,126).
5. Team Performance Ranking	Horizontal Bar Chart	Constructor Points in race_results: 1. Red Bull Racing (8,006), 2. Mercedes (7,790), 3. Ferrari (5,778), 4. McLaren (4,270), 5. Alpine (1,947), 6. Aston Martin (1,181), 7. Williams (1,057), 8. RB (923), 9. Haas (591), 10. Kick Sauber (363).
6. Team Championship Performance Analysis	Grouped Vertical Bar Chart	Built on championship_standings: Mercedes (118 wins, 7,790 pts, 7 titles), Red Bull Racing (100 wins, 8,006 pts, 7 titles), Ferrari (46 wins, 6,390 pts), McLaren (35 wins, 4,270 pts, 3 titles).
7. Tyre Compound Usage Distribution	Pie Chart	Starting compound distribution across 6,464 race results: Soft (1,889 | 29.22%), Medium (1,884 | 29.15%), Hard (1,849 | 28.60%), Intermediate (842 | 13.03%).
8. Country-wise Race Participation	Treemap / Highlight Table	Hierarchical country blocks sized by race entries: Bahrain (2,168 | 33.54%), USA (400), Monaco (328), Hungary (328), Spain (328), Canada (308), UAE (308), Brazil (308), Singapore (288).
9. Podium Achievements by Driver and Team	Packed Bubble Chart	1,002 total podiums sized by driver podiums & colored by team: Max Verstappen (187), Lewis Hamilton (174), George Russell (115), Charles Leclerc (113), Sergio Perez (97), Lando Norris (85), Carlos Sainz (63), Oscar Piastri (35), Esteban Ocon (25).

4. 9-Scene Data Story & Flask Web Integration (To-Do 6, To-Do 11, To-Do 12)

The data story — **Formula 1 Race Intelligence: From Global Overview to Championship Success** — is structured into **9 sequential scenes** matching each analytical dimension: (1) Global Scale Overview (KPIs), (2) Geographic Driver Distribution, (3) Driver Reliability & DNF Analysis, (4) Driver Performance Leaderboard, (5) Team Performance Ranking, (6) Season-by-Season Championship Analysis, (7) Tyre Strategy Patterns, (8) Country-Level Race Participation, and (9) Podium Achievement Distribution across Drivers and Teams.

Web Integration (Flask + GitHub Pages): The published Tableau Public workbook and interactive Chart.js/SVG replica are integrated into a responsive dark-themed web application (**web/app.py** and **index.html**), supporting full interactive filtering across DNF Reason, Year of Race Date (2010–2026), and the Top N Parameter.